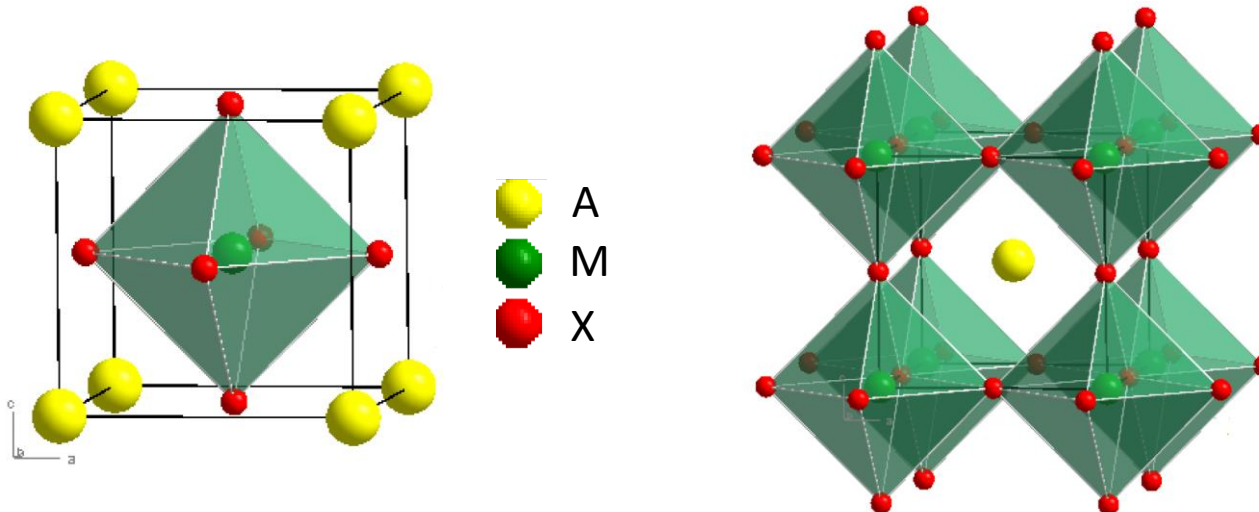


Fundamentals of Photovoltaics (EE336)

Lecture 8 – Perovskite solar cell

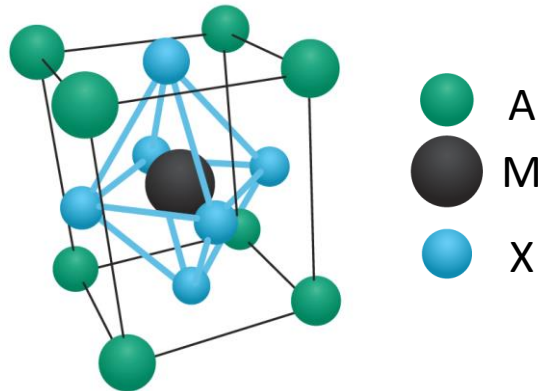
What is perovskite?

- Perovskite is the common nomenclature for compounds with the generic chemical formula – AMX_3 .
- In the cubic unit cell, the A-cation resides at the eight corners of the cube, while M-cations is located at the body center that is surrounded by 6 X-anions (located at the face centers) in an octahedral $[MX_6]^{4-}$ cluster.
- Typical inorganic perovskites are $CaTiO_3$ and $SrTiO_3$.



Note: $CaTiO_3$ mineral was discovered in the Ural mountains (Russia) in 1839 and is named after Russian mineralogist L.A. Perovski (1792–1856)

- The compounds being used in the perovskite solar cell (PSC) belong to the family of **organic-inorganic metal halide (OIMH) perovskite**, in which the organic cations are located at A sites, the metal cations are located at M sites and halides are at X.

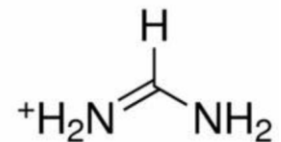
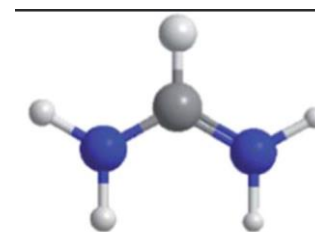
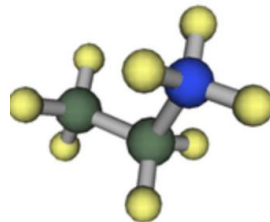
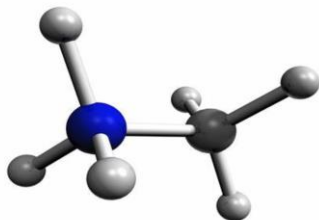


A = organic cations (methylammonium (MA) CH_3NH_3^+ , ethylammonium (EA) $\text{C}_2\text{H}_5\text{NH}_3^+$, formamidinium (FA) etc.)

M = metal cations (Pb^{2+} , Sn^{2+} , Ge^{2+} , Cu^{2+} etc.)

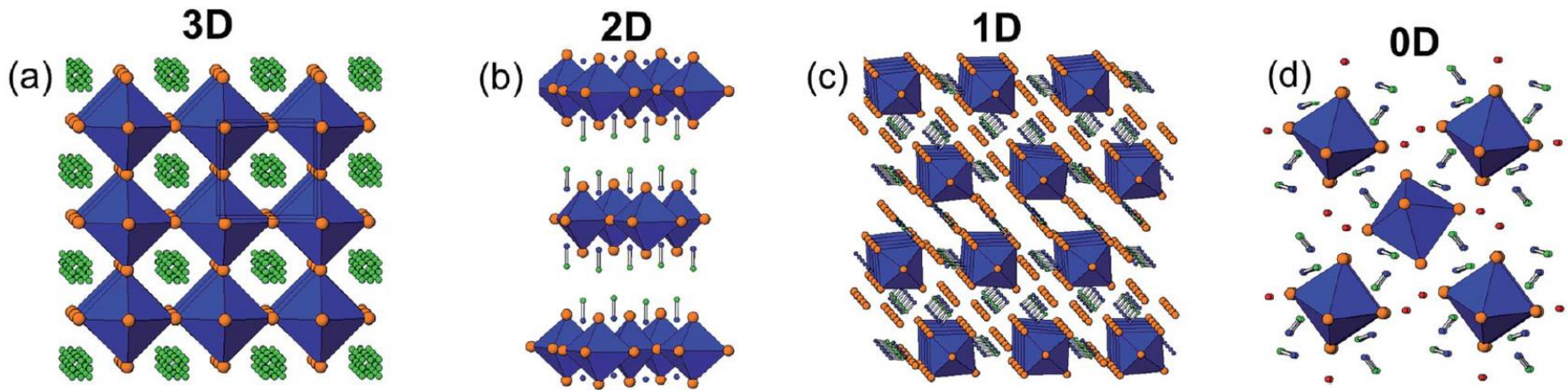
X = halide anions (I^- , Cl^- and Br^-)

- Methylammonium lead iodide (MAPbI_3) ($\text{CH}_3\text{NH}_3\text{PbI}_3$) is an OIMH perovskite commonly used in PSC.
- In $\text{CH}_3\text{NH}_3\text{PbI}_3$, each octahedral $[\text{PbI}_6]^{4-}$ is connected with six neighbours at the iodide – forming a **3-D network**. The countercation (CH_3NH_3^+) is located at the void of the network.



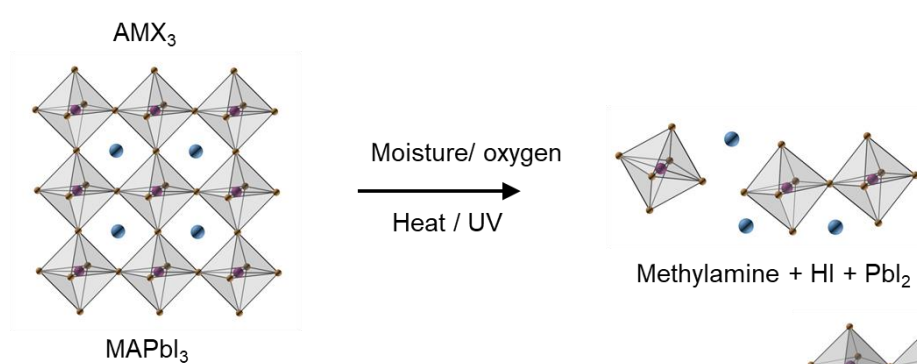
Formamidinium

- For the **2-D case** (e.g., $(\text{CH}_3\text{NH}_3)_2\text{PbI}_4$), each $[\text{PbI}_6]_4$ octahedron is connected with four neighbours at the halide – forming a 2-D network layer that is sandwiched between two CH_3NH_3^+ layers. A multiple quantum well structure results from the stacking of these sandwiched layers via van der Waals interaction – with the 2-D $[\text{PbI}_6]_4$ functioning as the **quantum well** and the CH_3NH_3^+ layer as **the barrier**.
- For the **1-D case** (e.g., $(\text{C}_{10}\text{H}_{21}\text{NH}_3)_2\text{PbI}_4$), each octahedron is connected at two opposite corners with its neighbour – forming separate infinite chains.
- The **quantum dot (0-D)** analogue manifests as $(\text{CH}_3\text{NH}_3)_4\text{PbI}_6 \cdot 2\text{H}_2\text{O}$, where each $[\text{PbI}_6]_4$ complex is neutralized by four CH_3NH_3^+ ions to form a molecule

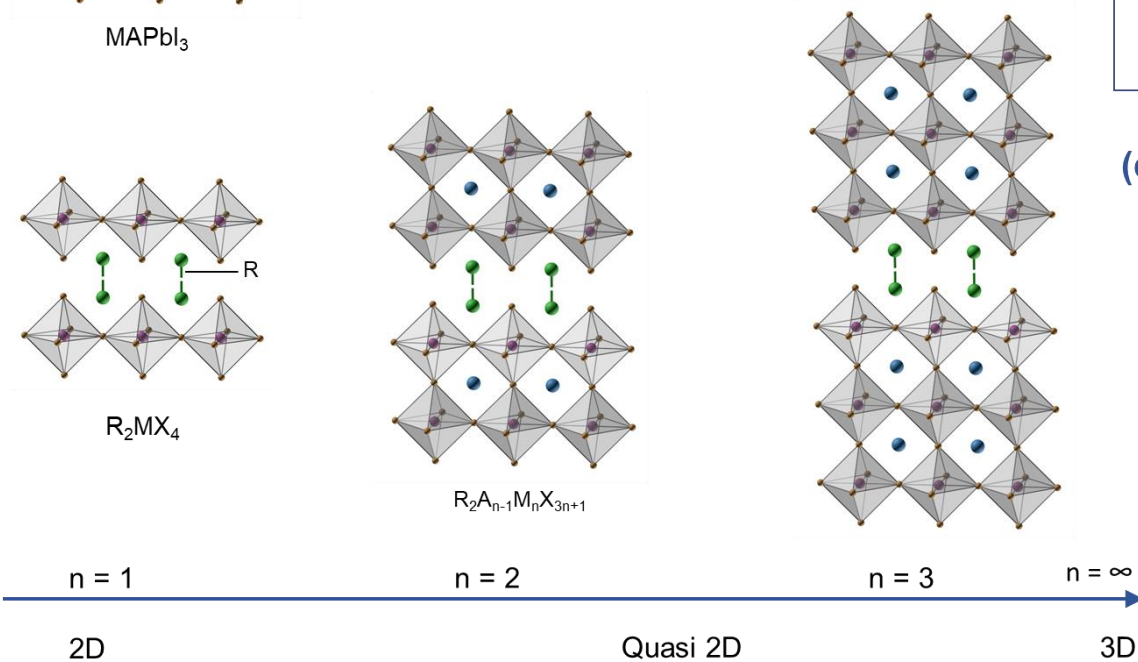


Ruddlesden-Popper 2D perovskite, General formula $R_2A_{n-1}M_nX_{3n+1}$

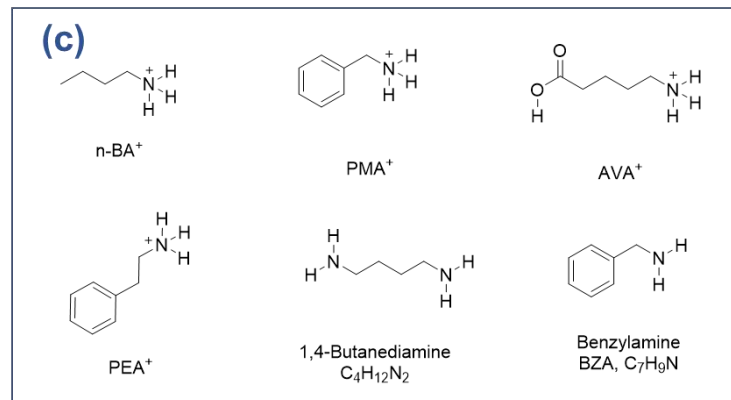
(a)



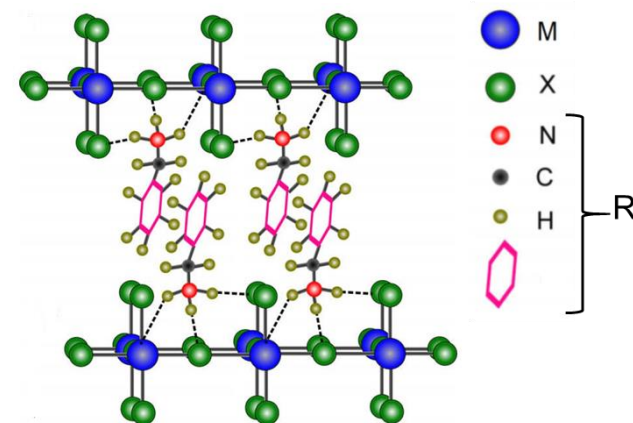
(b)



(c)



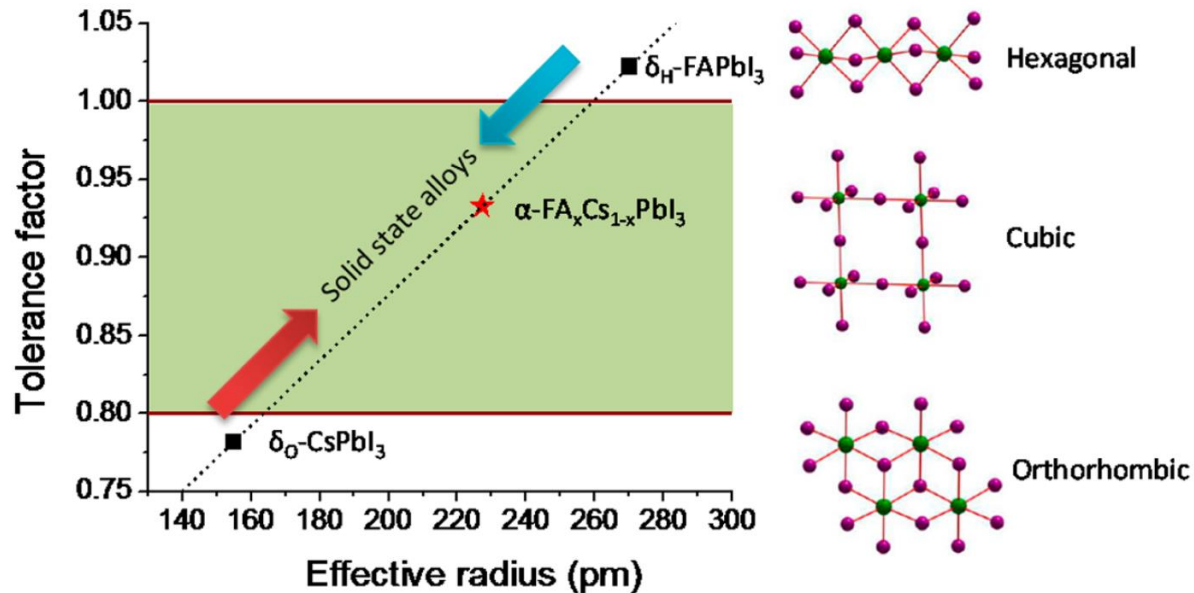
(d)



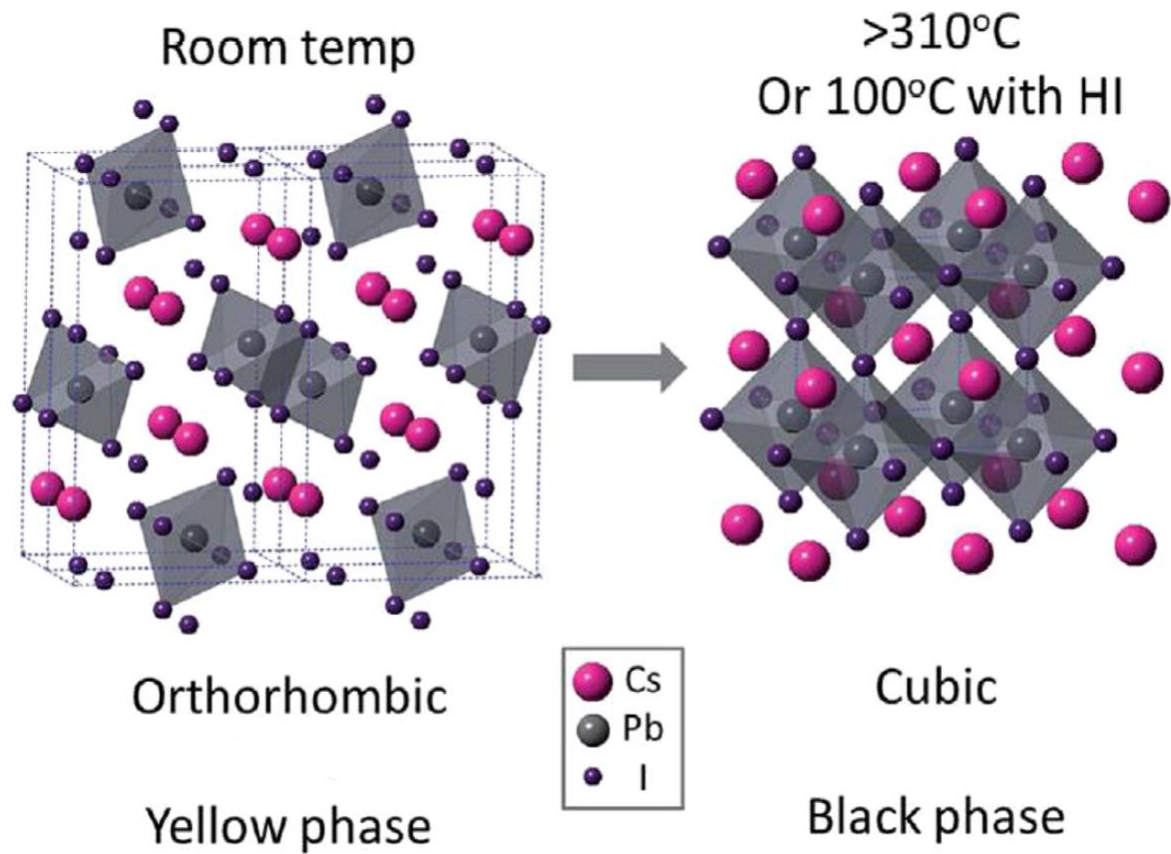
Tolerance Factor

- Material with AMX_3 composition can adapt different crystal structures depending on the size and interaction of the A cation and the corner-sharing MX_6 octahedra.
- Goldschmidt tolerance factor (t) is used to predict which structure is preferentially formed.
- The tolerance factor is calculated from the ionic radius of the atoms using the following expression.
- $$t = \frac{r_A + r_X}{\sqrt{2} (r_M + r_X)}$$
- where r_A is the radius of the A cation, r_M is the radius of M cation, and r_X is the radius of the anion.
- The tolerance factor assesses whether the A site cation can fit within the cavities in the MX_6 framework.
- A tolerance factor of 1 indicates a perfect fit and the perovskites can be formed only in the range $0.8 \leq t \leq 1$ although the lower part of this range they may be distorted due to tilting of the MX_6 octahedra.
- $t < 0.8$ indicates that the A cation is too small while $t > 1$ indicates that the A site cation is too large, and leading to the formation of nonperovskite structures (orthorhombic structure for $t < 0.8$ and hexagonal structure for $t > 1$).

- More than one structure is usually found for a perovskite material with a given chemical composition, depending on the temperature and preparation methods.
- For example, both MAPbCl_3 and MAPbBr_3 are found in a cubic structure at the room temperature with transitions to a tetragonal structure at lower temperatures (-119°C for MAPbCl_3 and -62°C for MAPbBr_3)

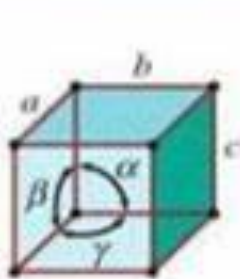


The α -phase denotes the phase with cubic structure and δ -phase denotes the nonperovskite structure



Quick review on different crystal structures

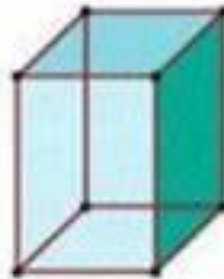
- There are seven unique arrangements, known as crystal systems, which fill in a three dimensional space.



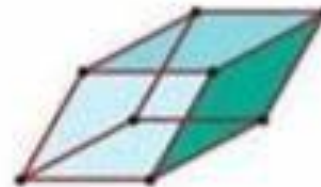
Simple cubic
 $a = b = c$
 $\alpha = \beta = \gamma = 90^\circ$



Tetragonal
 $a = b \neq c$
 $\alpha = \beta = \gamma = 90^\circ$



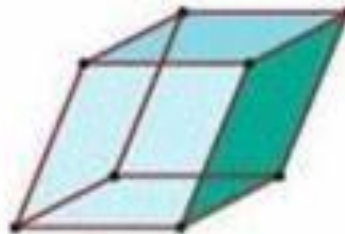
Orthorhombic
 $a \neq b \neq c$
 $\alpha = \beta = \gamma = 90^\circ$



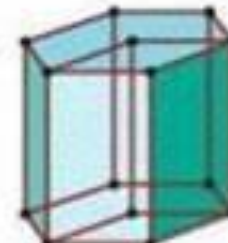
Rhombohedral
 $a = b = c$
 $\alpha = \beta = \gamma \neq 90^\circ$



Monoclinic
 $a \neq b \neq c$
 $\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$



Triclinic
 $a \neq b \neq c$
 $\alpha \neq \beta \neq \gamma \neq 90^\circ$



Hexagonal
 $a = b \neq c$
 $\alpha = \beta = 90^\circ, \gamma = 120^\circ$

Composition of ions and Bandgap tuning

The bandgap of the OIMH Perovskites can be tuned (and hence their optoelectronic properties) by changing the composition of ions.

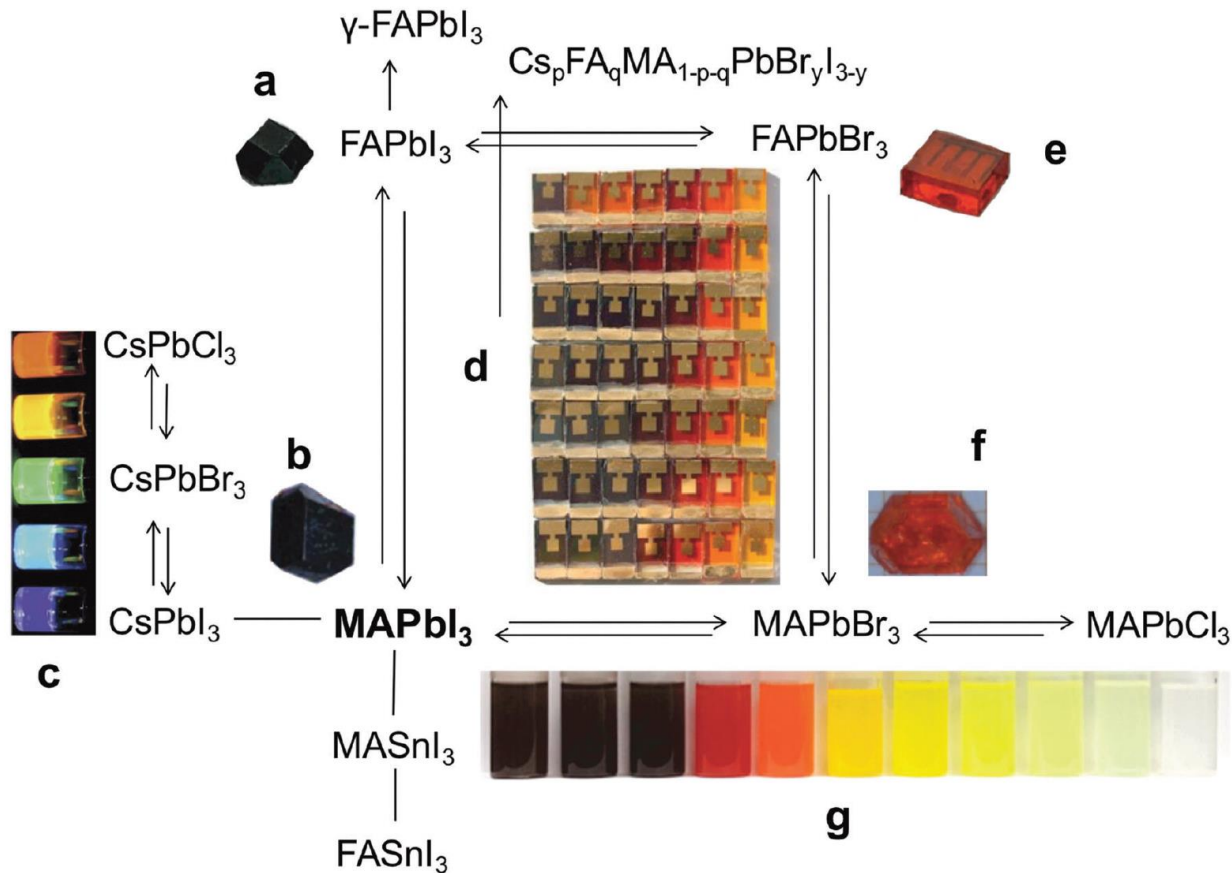
- Halide substitution

- ☐ The iodine in MAPbI_3 can be replaced with both chlorine and bromine.
- ☐ While the ionic size of halide decreases, the bandgap increases.
- ☐ For single crystal, the bandgap is 2.97, 2.24, and 1.53 eV for the Cl, Br and I perovskite respectively.
- ☐ For polycrystalline film, the bandgap becomes slightly higher, 3.1, 2.3 and 1.6 for the Cl, Br and I perovskite respectively.
- ☐ From the photovoltaic perspective, MAPbI_3 is suitable for single junction cell and MAPbBr_3 could be interesting for tandem application whereas MAPbCl_3 is relevant to light emitting devices.

- Monovalent Organic cation substitution

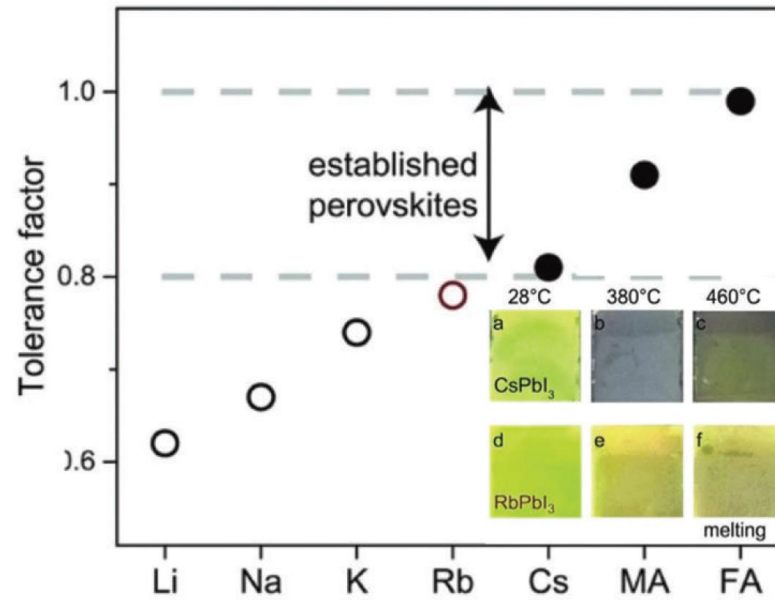
- ☐ The cation also can be replaced. MA can be replaced with slightly larger formamidinium ion (FA).
- ☐ However, the cation exchange only has a small impact on the optical bandgap.
- ☐ The larger FA ion expands the lattice, and change the tilt of the PbI_6 octahedra, which results in a slight decrease of the bandgap from ~ 1.59 eV for MAPbI_3 to ~ 1.45 - 1.52 for FAPbI_3 .
- ☐ From the perspective of light absorption, this is more suitable bandgap for a single junction cell.

- ❑ At room temperature, FAPbI_3 is found in cubic or pseudocubic (tetragonal structure very close to cubic).
- ❑ However, the cubic or pseudocubic α -phase at room temperature easily transforms into a non-photoactive phase (yellow phase) with hexagonal structure, which is not suitable for photovoltaic application.



- Organic/inorganic ion mixing

- ❑ Using perovskites with mixed cations is an important approach because the perovskite compounds with single organic or inorganic ion come with several disadvantages.
- ❑ For example, MAPbI_3 has never reached stabilized efficiencies beyond 20%. On the other hand, FAPbI_3 and CsPbI_3 are not stable in the cubic (or pseudocubic) α -phase at room temperature although FA appears to give advantages over MA.
- ❑ Therefore, it has become an important design principle to mix two or more cations to achieve perovskite compounds combining the advantages of the constituents while avoiding their drawbacks.
- ❑ In double cations MA/FA mixtures, a small amount of MA includes a preferable crystallization of FA perovskite into its photoactive black phase, resulting in more thermally and structurally stable composition than pure MA or FA compounds.
- ❑ However, photoinactive yellow phase impurities are present in MA/FA mixture.
- ❑ Therefore, adding Cs into MA/FA mixtures and forming triple cation perovskites (Cs/MA/FA) results in a substantially suppressed yellow phase.



Optoelectronic properties

Light absorption

- OIMH Perovskites have **high absorption coefficient**.
- For example, the absorption coefficient of MAPbI_3 ($> 3 \times 10^4 \text{ cm}^{-1}$ in the visible light region) is more than one order of magnitude larger than that of silicon.
- The strong absorption of MAPbI_3 can be attribute to its **direct bandgap** and Pb 6p- I 5p transition.
- They also exhibit **photon recycling property** in which photons are reabsorbed and re-emitted many times before an electron-hole pair is collected or a luminescent photon escapes. Previously, this kind of photon recycling property is only observed in GaAs, which gives the highest efficiency among different photovoltaic materials.

Charge generation and exciton binding energy

- The exciton binding energy of perovskite is quite low ($\sim 75 \text{ meV}$ for MAPbI_3 and $\sim 80 \text{ meV}$ for MAPbI_3). These values are much lower than exciton binding energy of organic materials ($0.2 - 1.0 \text{ eV}$).
- Therefore, unlike in OPV, the free charges can be generated without the energy offset at the heterojunction interface.
- The direct generation of free charges dominates the conversion process from photons to photocurrent in PSC.
- OIMH perovskite is only **non-excitonic material** among the solution-processed materials for photovoltaic application.

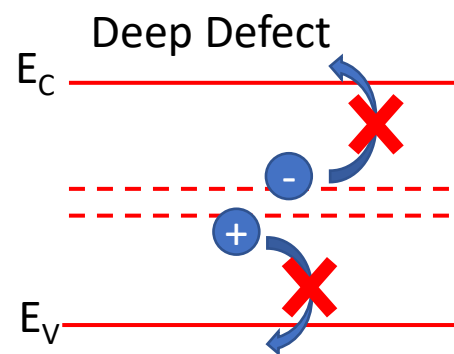
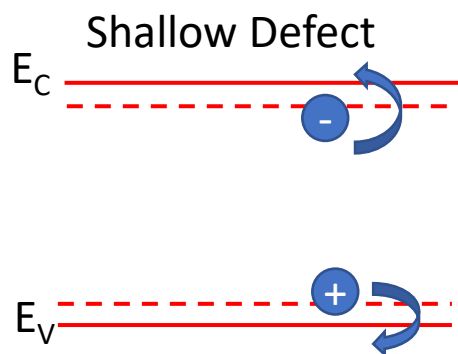
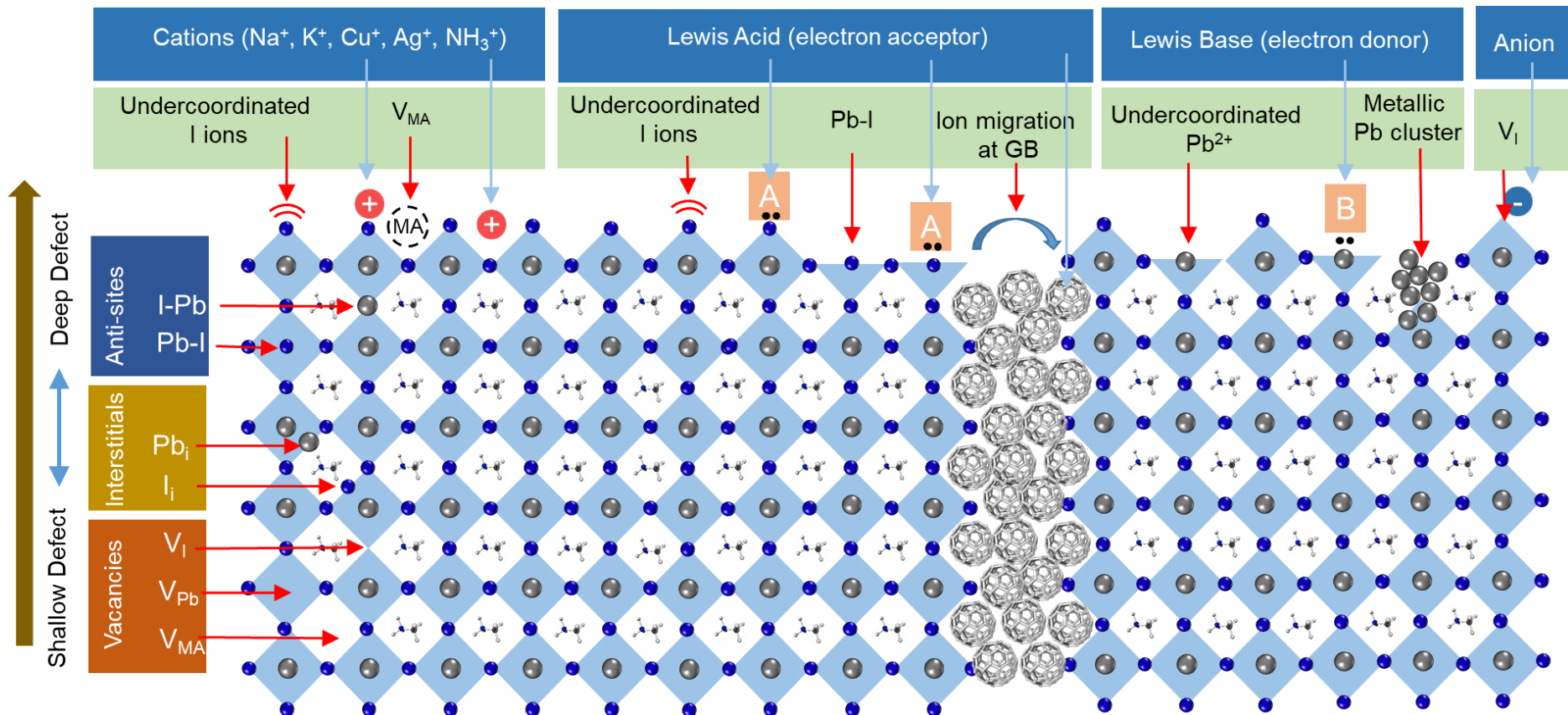
- The exciton binding energy in perovskites is also found to be sensitive to their **microstructure**.
- The excitonic features in MAPbI_3 are found in samples with large grain size of $\sim 1 \mu\text{m}$.
- In perovskite thin films with small grains, disorders and variations of electrostatic potential decrease the exciton binding energy and suppress exciton formation.
- Excitons are more likely generated at locations with higher crystallinity and free charges are found at the center of grains with many small domains.
- The **dimensionality** of the crystal structure also influences exciton binding energies.
- The exciton binding energy in 2D structure is higher than 3D structure owing to the spatial quantum confinement (quantum well). (Note: 2D structure has a better moisture stability than 3D structure although a high exciton binding energy and hence lower efficiency)
- It was proposed that excitons in 2D perovskites dissociate into long-lived free carriers with the assistance of **edge states**.

Charge transport and recombination

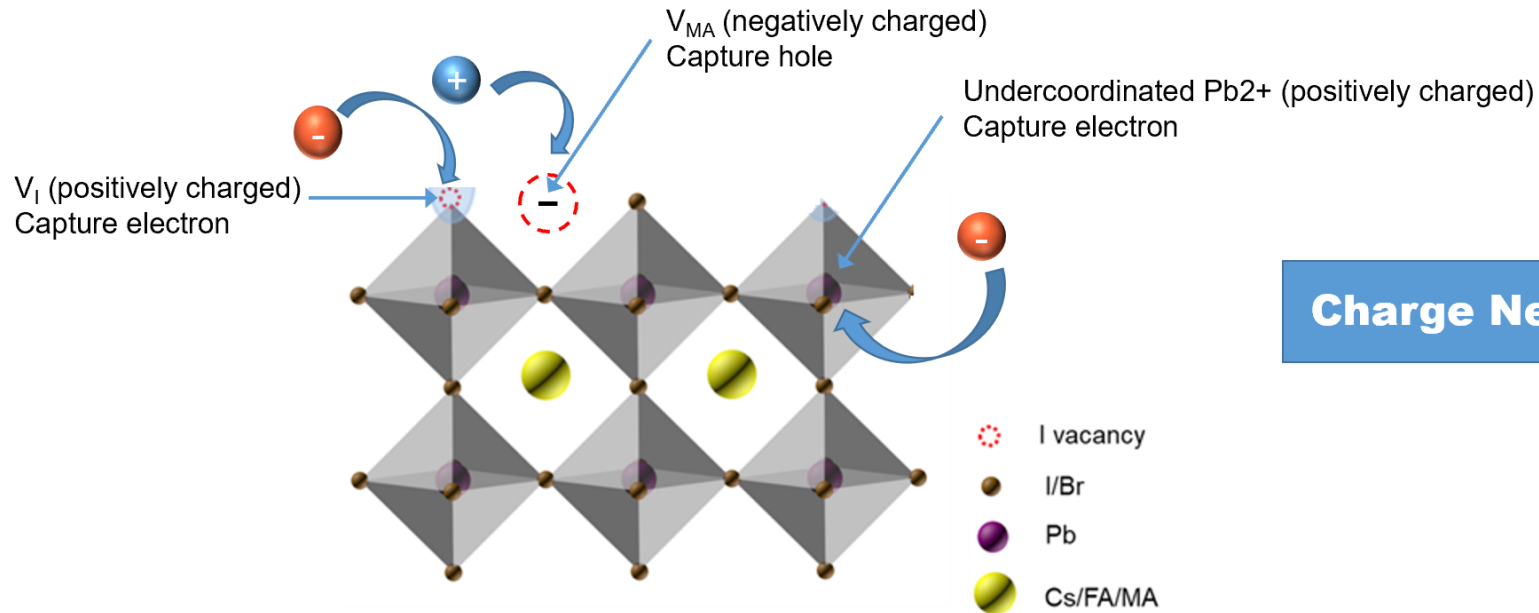
- The diffusion length in MAPbI_3 is about 100 nm for both electron and hole. Although this value is much lower than the diffusion length of carriers in classical inorganic semiconductor, it is relatively high compared to other low-temperature solution-processed films.
- The diffusion length in the mixed perovskite $\text{MAPbI}_{3-x}\text{Cl}_x$ exceeds 1 μm .
- Because of long diffusion length for both electron and hole, the charge transport in OIMH perovskite can be considered as ambipolar.
- The photoluminescence lifetime in single crystal perovskite is above 10 μs . The photoluminescence lifetime even in surface-passivated polycrystalline films can be as long as several microseconds.
- The recombination lifetime of polycrystalline film used in the perovskite solar cell can be as long as hundreds of nanosecond, which is considerably long for low-temperature solution-processed material.

Crystallization Energy and Defect Density

- The perovskite has **very low crystallization activation energy** of 57 – 97 kJmol⁻¹, depending on the lead salts used as precursors. By contrast, the crystallization activation energy of amorphous silicon can be as high as ~ 470 kJmol⁻¹.
- This low crystallization energy barrier leads to high crystallinity and low defect density even at low temperature processing.
- The **defect tolerance** of perovskite is also responsible for the long carrier recombination lifetime in polycrystalline film.
- It is now generally widely accepted that the low density of charge traps (defects) contributes to the large V_{oc} .
- Most of **the point defects that exist in the bulk of MAPbI₃** either have energy levels in the conduction band or valance band or form **relatively shallow traps**.
- Despite the point-defect tolerance of OIMH perovskites, the presence of extended defects, such as grain boundaries and interfaces, increase the trap density.



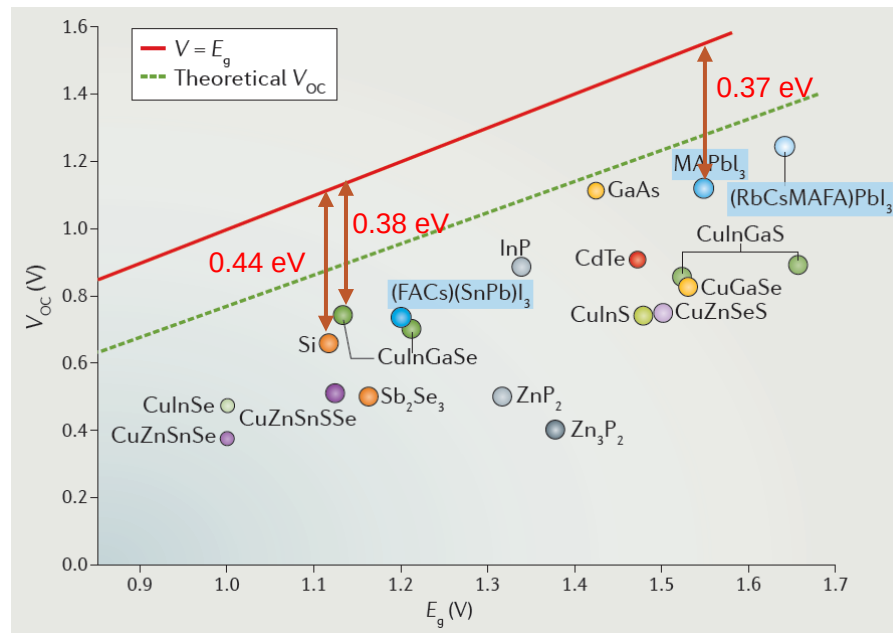
Defect Types and Passivation in Perovskite



Charge Neutrality

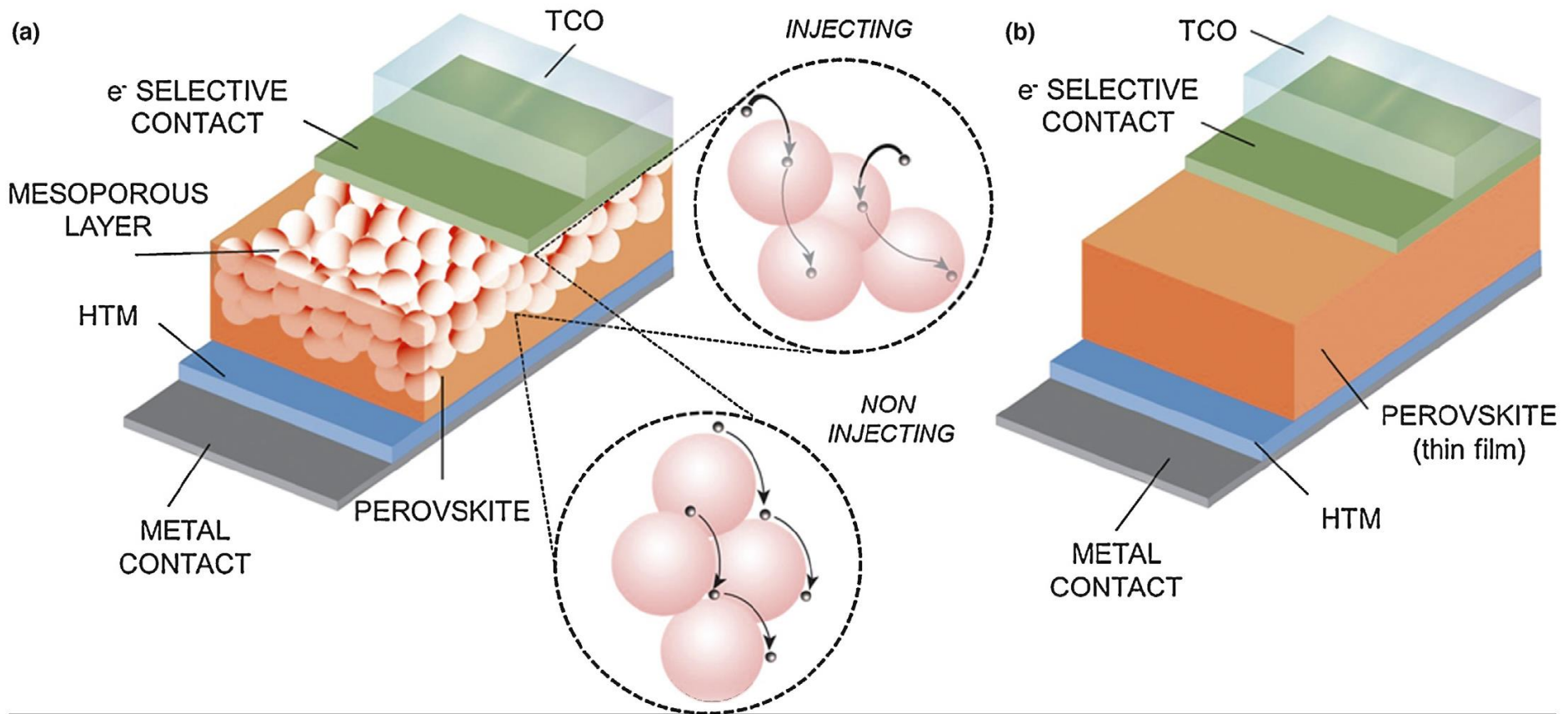
V_{OC} deficit in perovskite solar cell

- One significant feature of perovskite solar cell is the small V_{OC} deficit, which is defined by $E_g/q - V_{OC}$
- The V_{OC} deficit in the best perovskite solar cell (~ 0.37 eV) is much smaller than that of organic (> 0.6 eV) and quantum dot solar cells, is even smaller than that of inorganic c-Si (0.44 eV) and close to that of CIGS solar cell (0.38 eV).

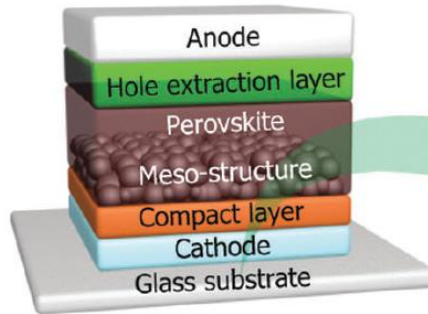


Device Architectures and Evolution

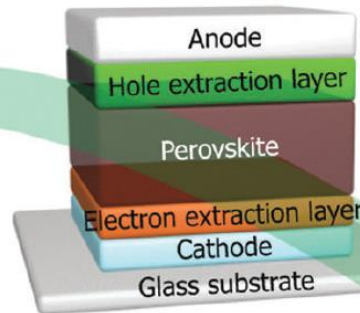
- The pioneered perovskite solar cells are fabricated in DSC structure (with liquid electrolyte) in which dye sensitizer is replaced with perovskite absorber.
- The first perovskite solar cell reported in 2009 show efficiency of 3.81% using MAPbI_3 .
- The second reported perovskite solar cell reported in 2011 yield efficiency of 6.5% using MAPbI_3 nanocrystal (quantum dot) .
- However, the stability of pioneered device is very poor due to the dissolution of halide in liquid electrolyte.
- This problem was solved by using a solid state hole transporting layer (Spiro-OMeTAD), on top of the perovskite layer instead of using the liquid electrolyte. This structure is usually called **mesoporous structure**. Today, the perovskite solar cell with mesoporous structure yields the efficiency more than 25%.
- The semiconducting TiO_2 mesoporous layer can be replaced with an insulating Al_2O_3 scaffold.
- It is worth to remark that regardless of the mesoporous layer, **a compact TiO_2 layer** is still required for both the collection of the generated electrons and hole blocking.
- Since Al_2O_3 's conduction band is far higher than the perovskite's LUMO, no electron injection from perovskite takes place. This indicates that the charge separation and electron transport occurs within perovskite.



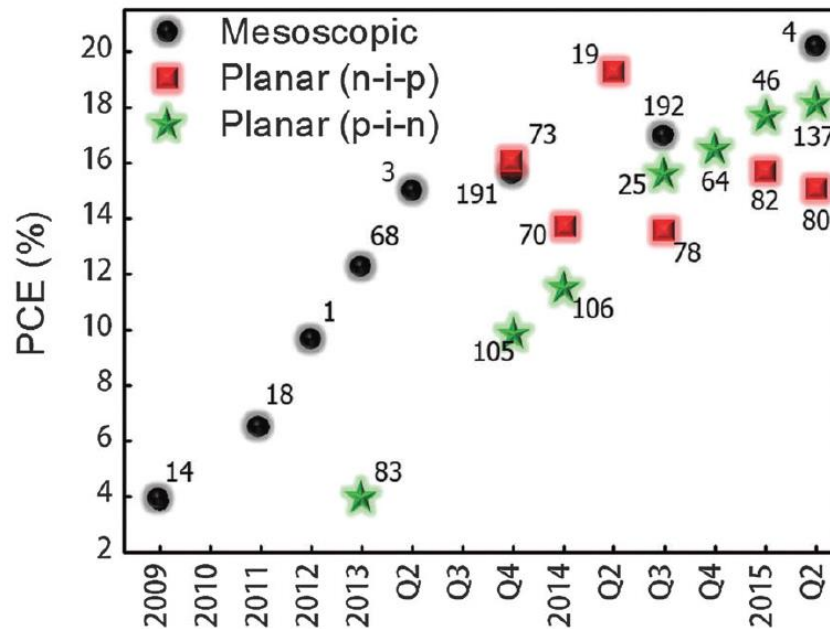
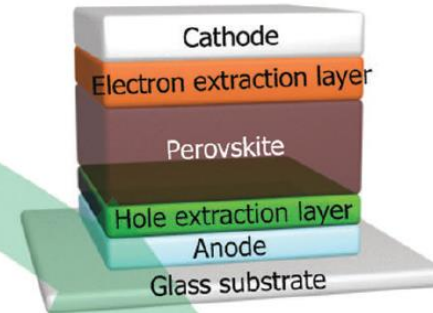
Mesoscopic PrSCs



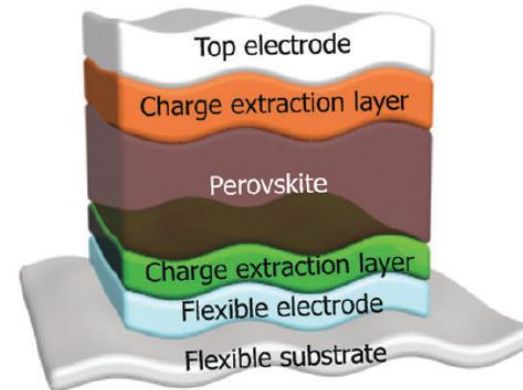
PHJ (n-i-p) PrSCs



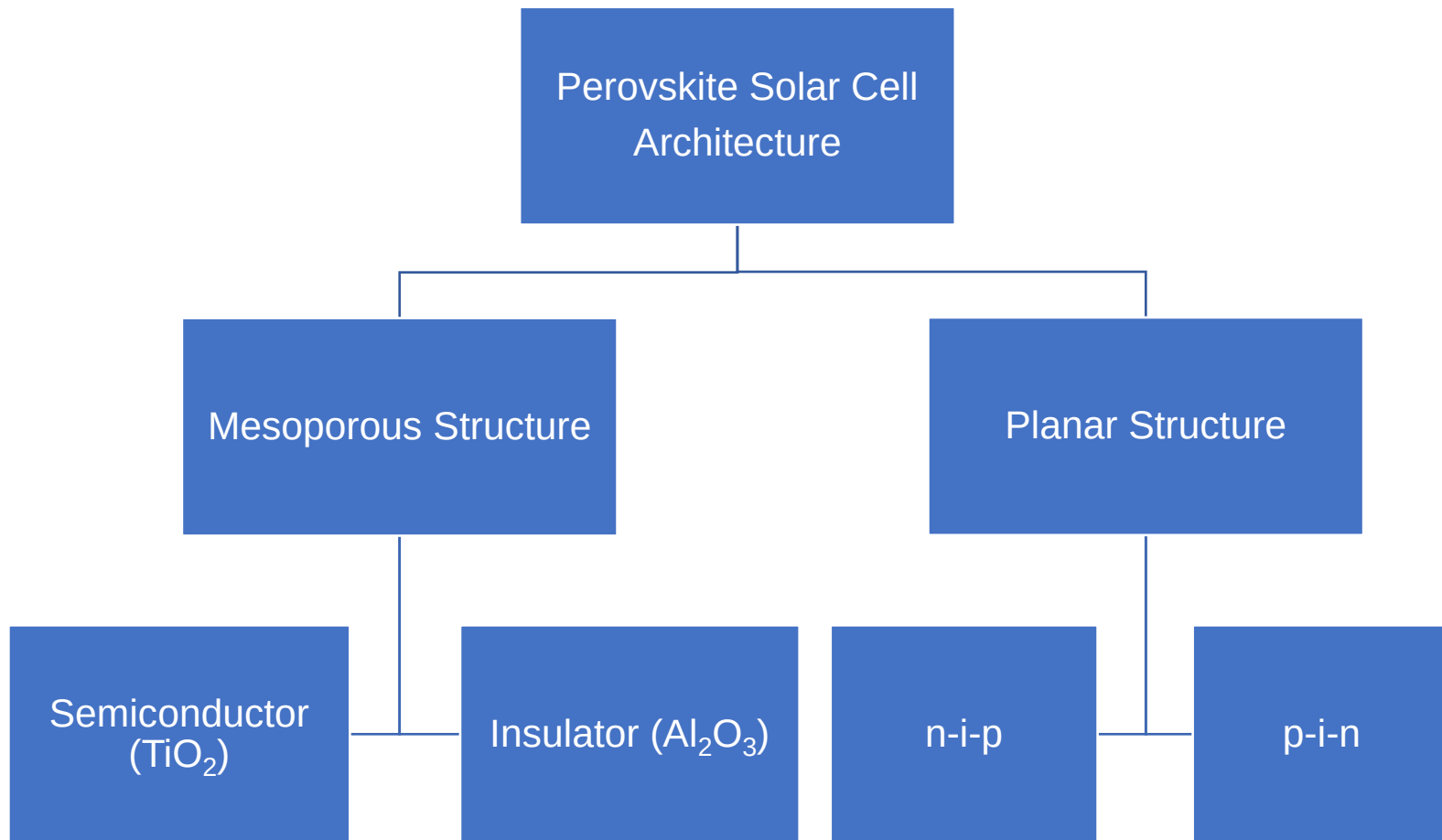
PHJ (p-i-n) PrSCs



Flexible PHJ PrSCs



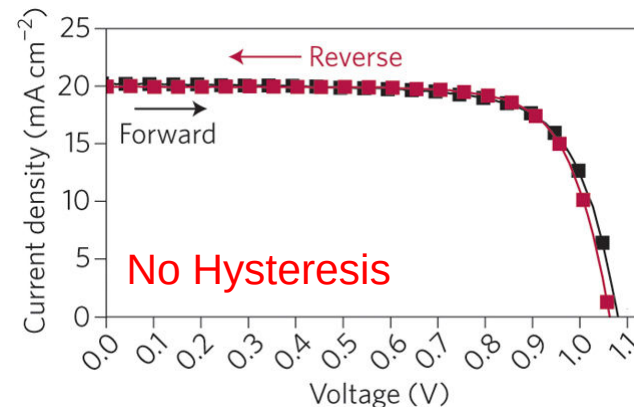
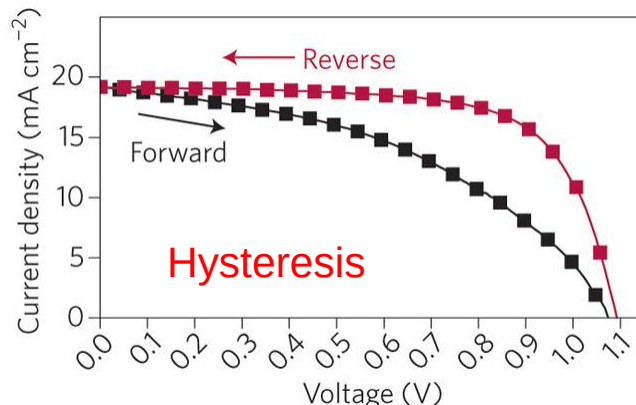
PHJ = planar heterojunction
Mesoscopic = Mesoporous



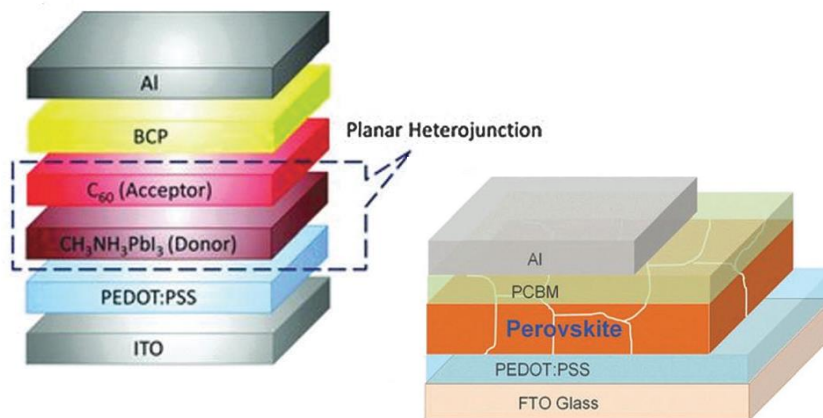
- Perovskite solar cell can also be fabricated with planar structure, without mesoporous layer.
- ~~Although the efficiency of planar structure is still lower than mesoporous structure, the former has advantage to make flexible solar cell due to the removal of mesoporous layer which is required high temperature annealing around 450 °C. (Note: so far, it is still not clear why mesoporous structure gives higher efficiency than planar structure)~~
- Planar structure is very similar to OPV structure; perovskite layer is analogous to BHJ of OPV.
- Planar structure can be divided into two groups according to the layer stacking sequence; the n-i-p structure and p-i-n structure.
- In n-i-p structure, the electron extraction/transport layer (EEL) is deposited on the bottom cathode, the hole extraction/transport layer (HEL/HTL) is underneath the top anode layer, and the perovskite is sandwiched between EEL and HEL/HTL.
- The p-i-n structure is the inverse of the n-i-p type.

Planar structure device and EEL

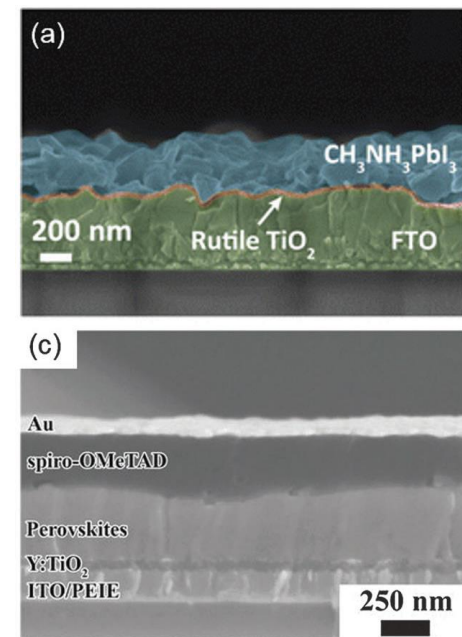
- In n-i-p planar structure device, the compact TiO_2 layer is generally used as EEL.
- The chemical composition and conditions of process and thermal treatment of TiO_2 film affect the performance of the device.
- SnO_2 EEL deposited by atomic layer deposition (ALD) or even solution-processed approach also yields a high efficiency.
- Other materials used as EEL are Yttrium-doped TiO_2 , TiO_x , ZnO (nanoparticles), CdSe nanocrystals, PCBM and C_{60} -SAM.
- In early days, the planar structure device suffers from hysteresis (non-identical I-V curves in forward and reverse scan) but it is suppressed or almost eliminated after careful optimization in perovskite layer and EEL.



- In p-i-n planar structure device, however, only PCBM is commonly used as EEL. Metal oxide EEL is not favourable in p-i-n structure (**Why?**)
- The solvents used for metal oxide are usually polar solvent such as ethanol, methanol, 2-methoxyethanol but perovskite is dissolved in polar solvent. On the other hand, the solvents used for PCBM such as chloroform, chlorobenzene, etc. are orthogonal solvent for perovskite and hence they do no damage the underlying film.
- C60 and BPC (Bathocuproine) are usually thermally evaporated on PCBM for a better performance.



Examples of p-i-n structured device



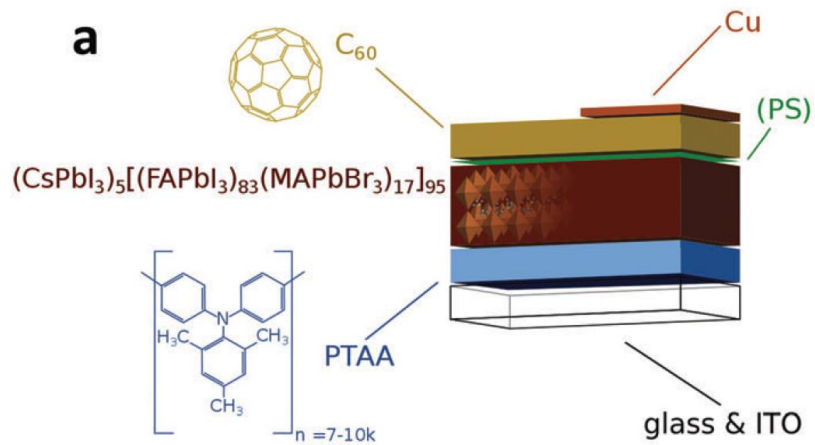
Examples of n-i-p structured device

Hole transport layer (HTL)

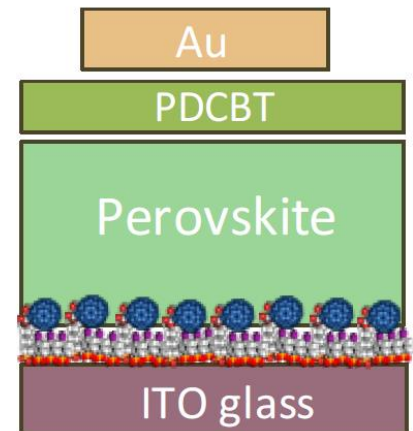
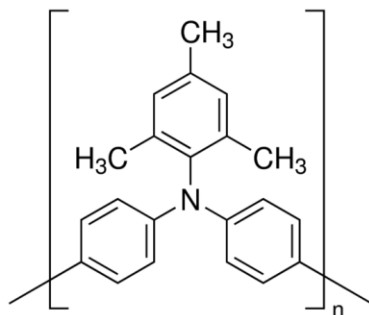
- HTL used in perovskite solar cell can be classified into 4 categories: small molecule, polymer, carbon and inorganic.
- Small molecule HTL
 - ❑ Small molecules offer the advantages of flexible processing from both solution and evaporation.
 - ❑ The most effective small molecule HTL to date are spiro-based triphenylamine derivatives (e.g. spiro-MeOTAD).
 - ❑ These molecular backbones have a relatively low tendency to crystallize, which enables depositing amorphous organic layers conformally and thus with a good electronic contact to the perovskite surface.
 - ❑ Chemical **doping of the small molecule HTL is an essential step** to prepare highly efficient cell. Lithium salts, organic Lewis bases and metal – organic oxidants are used as dopant.
 - ❑ Increasing the hole transport capabilities (conductivity) while maintaining a low charge recombination at the interface with perovskite is the effect of doping the HTL.
 - ❑ However, **doping usually affects the long term stability of device.**
 - ❑ The small molecule HTLs are usually deposited on the perovskite film in mesoporous structure and n-i-p planar structure.

- Polymeric HTL

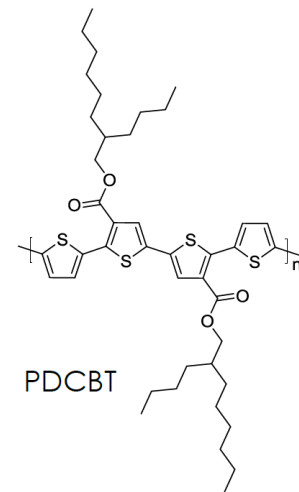
- ☐ Polymers are easy to form uniform and pin-hole free film.
- ☐ Extremely thin (10 nm) polymeric HTLs have been demonstrated to be effective for achieving a uniform coating on the perovskite film surface, whereas small molecules are needed above 100 nm thick layers to avoid pinholes.
- ☐ Polymeric HTLs can be exploited as both top and bottom selective contacts (i.e., in both p-i-n and n-i-p structures) while small molecule counterparts are usually used as top selective contact only (only in p-i-n or mesoporous structure) because they are relatively less stable to solvent exposure.
- ☐ High efficiency can be obtained without doping.
- ☐ A drawback, however, is impurities and batch-to-batch variation is higher in polymeric materials than small molecule. This will have impact on large scale applications of polymeric HTL.
- ☐ PTAA and PDCBT polymers render high efficiency comparable to spiro-MeOTAD (> 21%) and remarkable stability under full illumination.
- ☐ PEDOT:PSS is widely used as HTL in p-i-n structure but it is not suitable for n-i-p structure. (Why?) Its water absorption nature also affects the stability of the device.

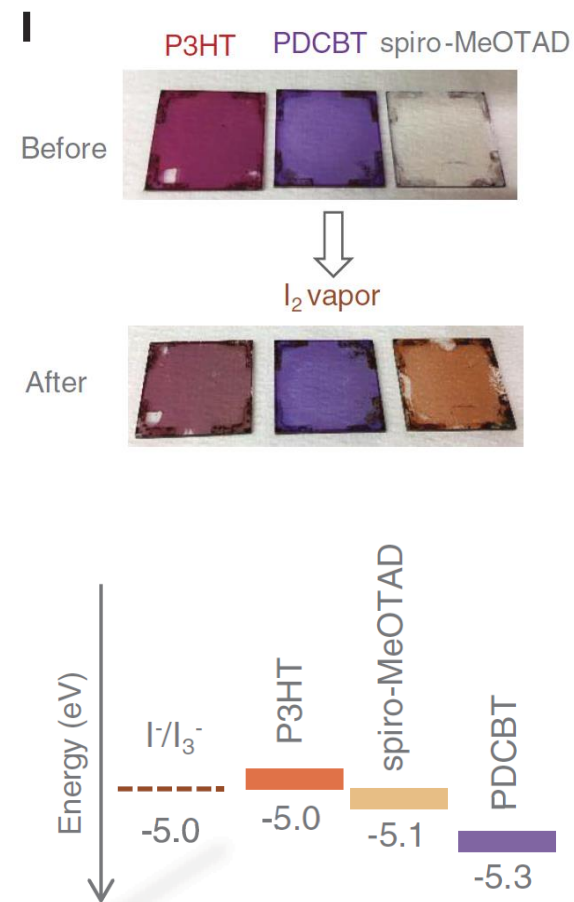
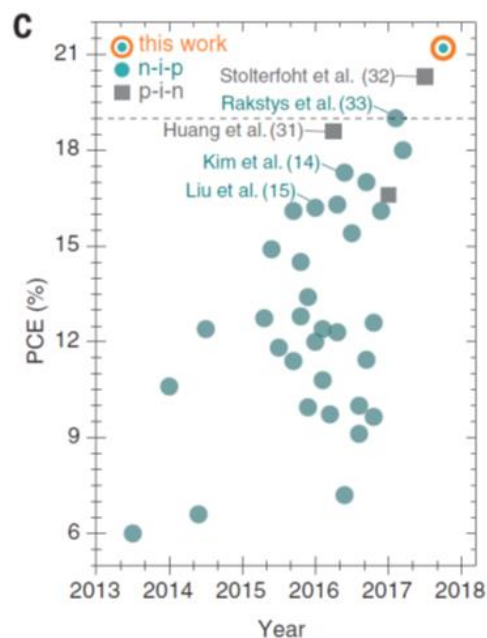
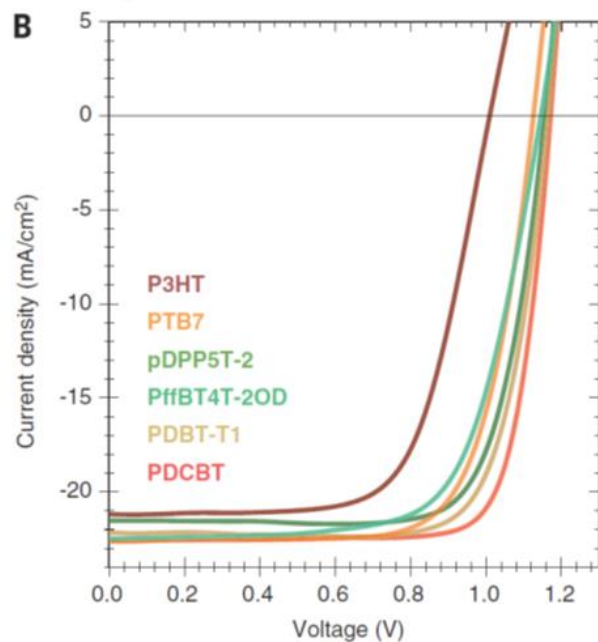
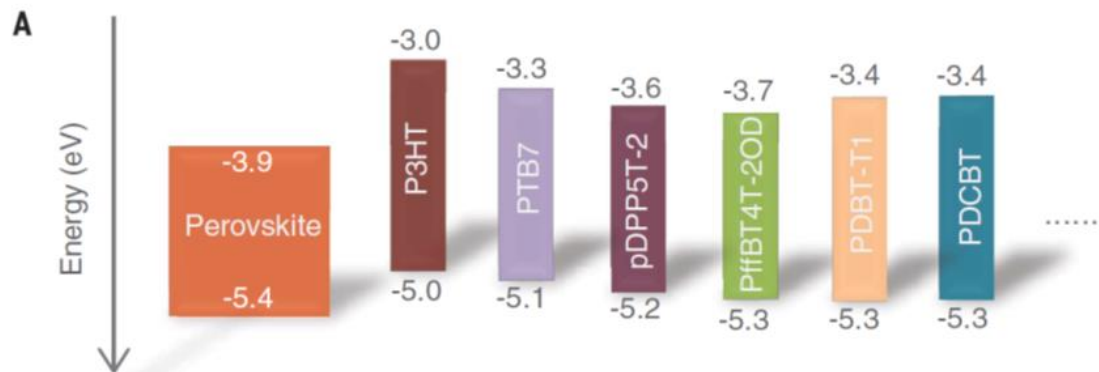


Polymeric HTL (PTAA) in p-i-n structure



Polymeric HTL (PDCBT) in n-i-p structure

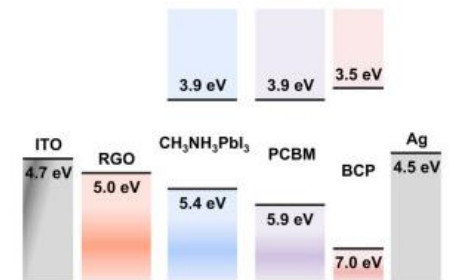
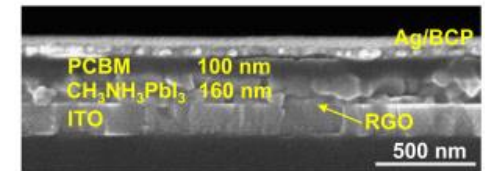
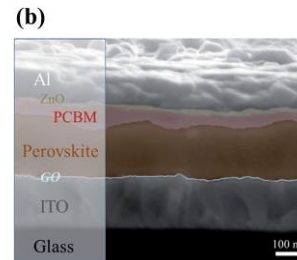
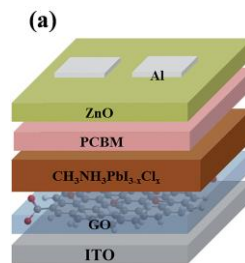
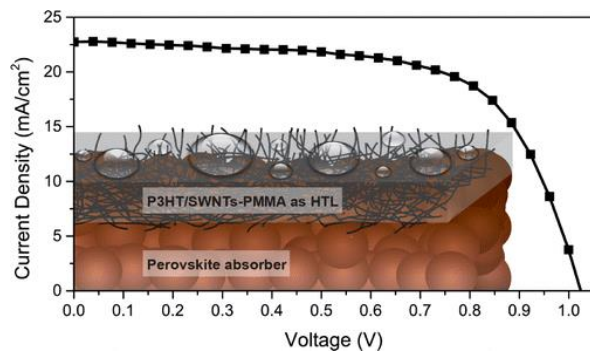




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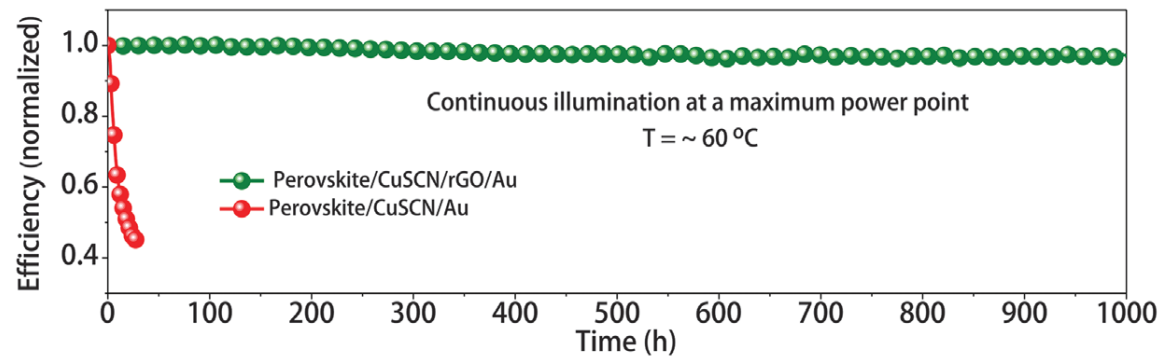
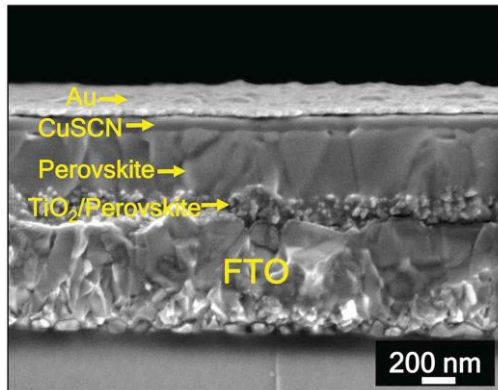
- Carbon HTL

- ❑ Carbon HTLs based on **nanotubes and nanopowders**, have been shown as stable alternatives to the organics.
- ❑ Carbon nanotube HTLs have been used to prepare highly efficient perovskite solar cells but they may require expensive purification procedures to isolate the right semiconductor tubes.
- ❑ The highest reported efficiency for a carbon nanotube, metal-free device, is 15.5%.
- ❑ Nanopowders offer one of the most economical HTL solution, which is compatible with the large-scale production lines.
- ❑ However highly efficient perovskite solar cells prepared with nanopowders HTLs are still lacking.
- ❑ **Graphene oxide (GO) and reduced graphene oxide (rGO)** are also used as HTL in p-i-n planar structure.



- Inorganic HTLs

- ❑ Among various inorganic HTLs, Copper(I) thiocyanate (CuSCN) is the most promising material.
- ❑ It is an extremely cheap, abundant p-type semiconductor that exhibits high hole mobility and a well-aligned work function with perovskite.
- ❑ The efficiency of more than 20% has been demonstrated in a mesoporous structure device without any dopant.
- ❑ The device stability is significantly improved by inserting a thin rGO between CuSCN and Au electrode.
- ❑ >95% of initial efficiency is retained after aging at a maximum power point for 1000 hours under full solar intensity at 60°C.
- ❑ Other inorganic materials such as MoO_3 and NiO_x are also used as HTL, usually in p-i-n planar structure.

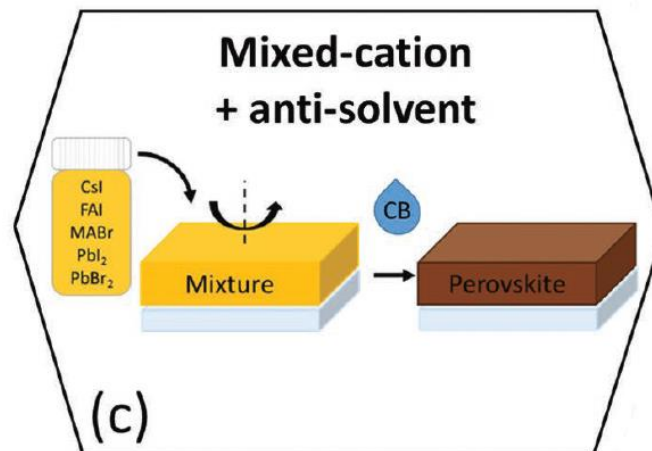
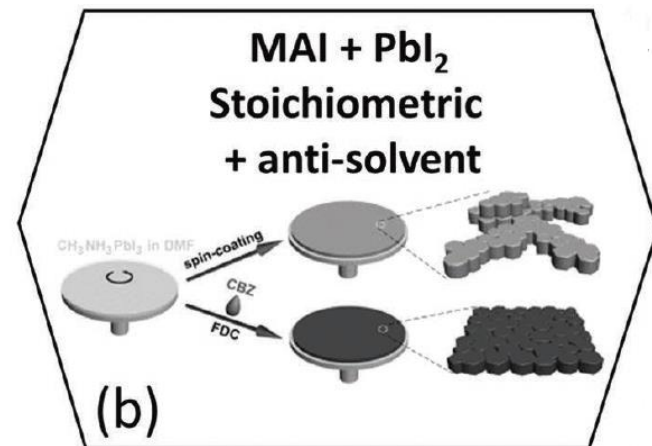
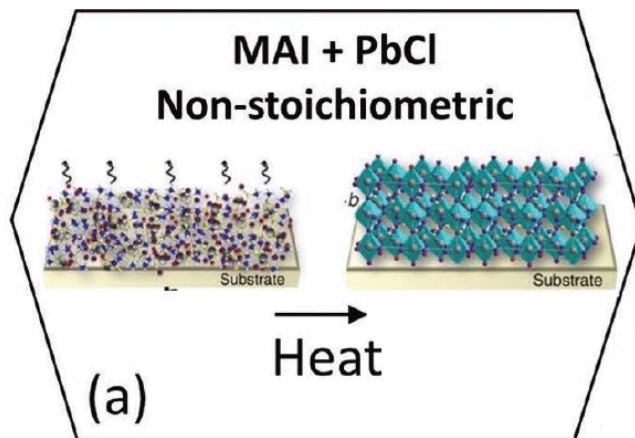


Crystallization of Perovskite

One-step deposition

- In one-step deposition, two precursors (methylammonium iodide (MAI) and lead (II) iodide (PbI_2)) are dissolved in specific ratio in polar solvent such as dimethylformamide (DMF), dimethyl sulfoxide (DMSO).
- The choice of solvent, precursors species (such as PbI_2 , PbCl_2 , PbAc_2) and their concentration have shown strong impact on the crystallization.
- One-step method often leads to inhomogeneous coverage of the film.
- **Adding an antisolvent** on top of the wet precursor film during spin-coating induces fast crystallization.
- The use of antisolvent promotes fast nucleation, which results in micron-sized crystals. It also leads to uniform and smooth perovskite films thus more reproducible devices.
- Some examples of antisolvent are chlorobenzene, benzene, toluene, isopropylalcohol (IPA) and chloroform.

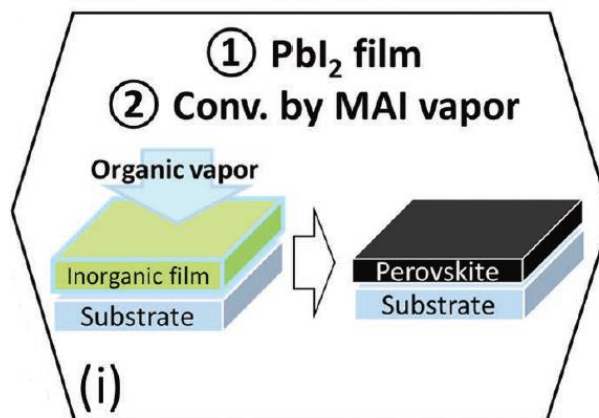
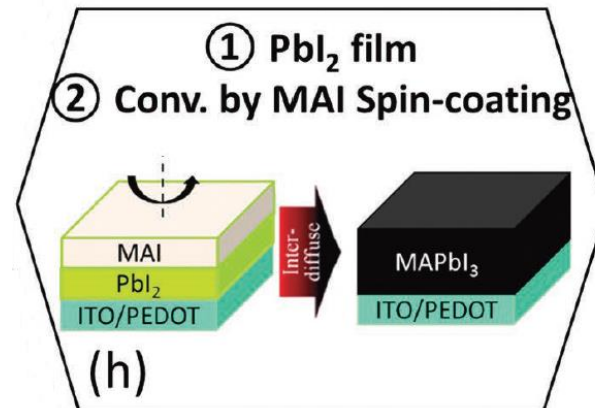
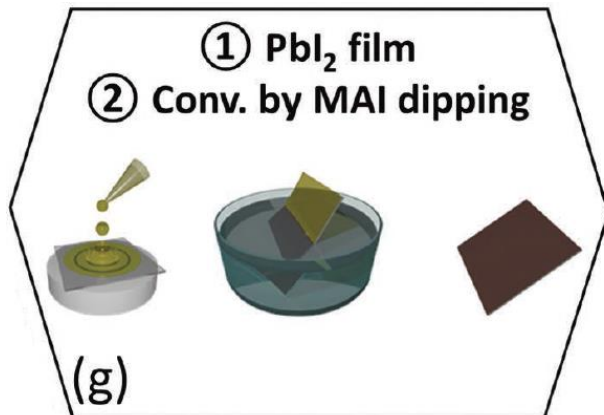
Examples of one-step deposition



Two-step deposition (Sequential deposition)

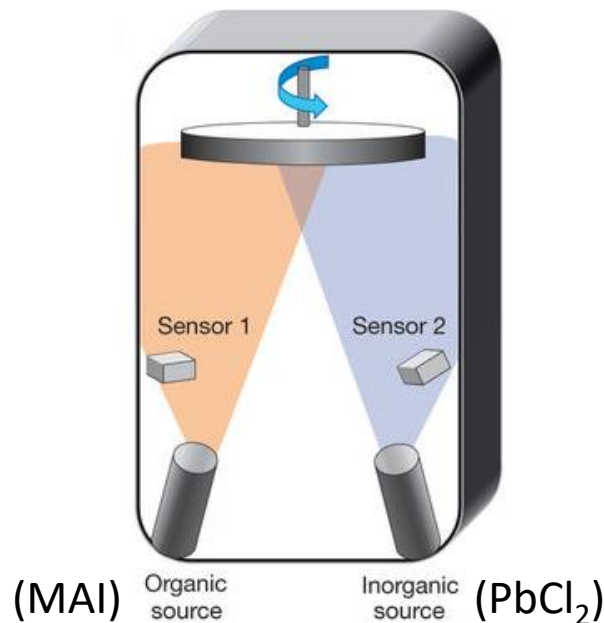
- Method based on conversion of pre-deposited PbI_2 into perovskite.
- PbI_2 solution is spin-coated onto mesoporous TiO_2 (first step) and subsequent immersion in a MAI-containing solution (second step).
- The key parameters to control the film quality for two-step deposition are immersion time, temperature of the conversion solution, and precursor concentration.
- Two mechanisms for transformation of PbI_2 to MAPbI_3 :
 1. in situ conversion: the original morphology of PbI_2 crystal is preserved while it is converted to MAPbI_3 via incorporation of MAI ions.
 2. dissolution-recrystallization mechanism: the crystal dissolves and the perovskite recrystallize in a thermodynamically favoured shape.
- These two mechanisms compete during crystallization, while in situ conversion is favoured in environments with a low MAI concentration.
- Other possibilities to convert pre-deposited PbI_2 into perovskite: by spin-coating concentrated MAI solution on the top of PbI_2 film, or by exposing the PbI_2 film to an MAI vapour atmosphere.
- Two-step method is also applicable in planar structured device. However, the film crystallization is inherently be different on planar substrate and different process optimization is required for planar structure.

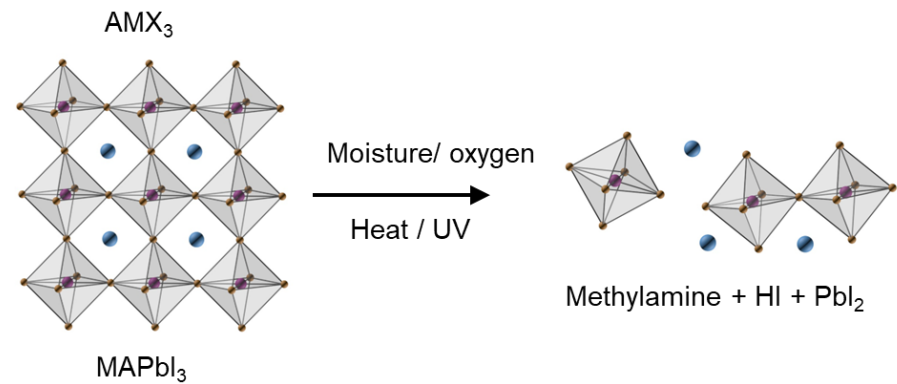
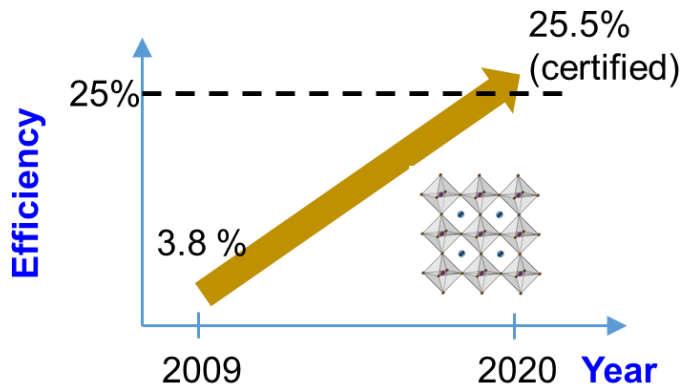
Examples of two-step deposition



Vacuum Deposition Method

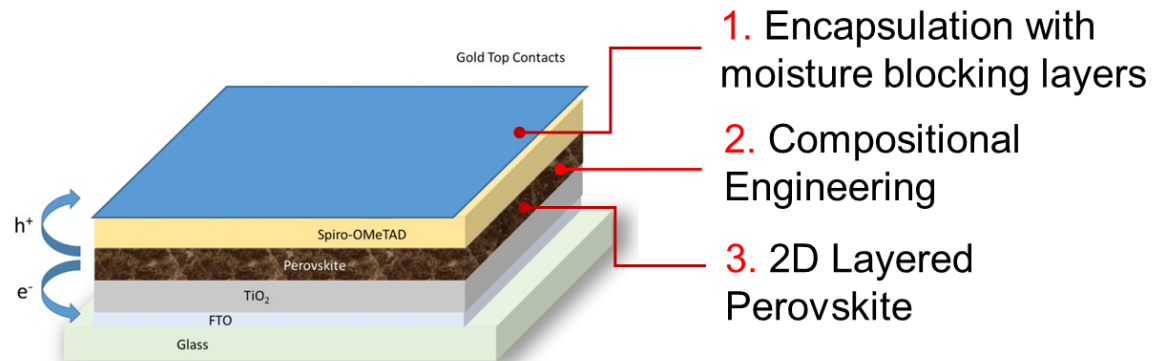
- In addition to the solution processed, perovskite crystal can be deposited by vacuum deposition method.
- It can be deposited through dual-source evaporation from two precursor salts simultaneously. For example, $\text{MAPbI}_{3-x}\text{Cl}_x$ can be deposited from MAI and PbCl_2 precursors. The molar ratio between two precursors can be tuned by controlling the evaporation rate of precursors.
- It is favourable for smooth, uniform, pin-hole free perovskite formation.





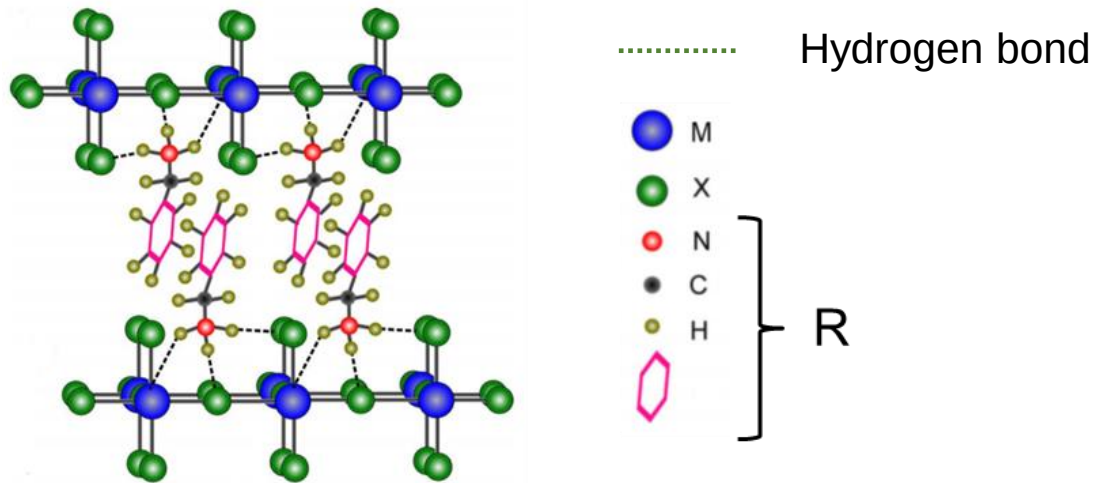
Technology Bottleneck: Stability

Current research status to tackle stability : ➡



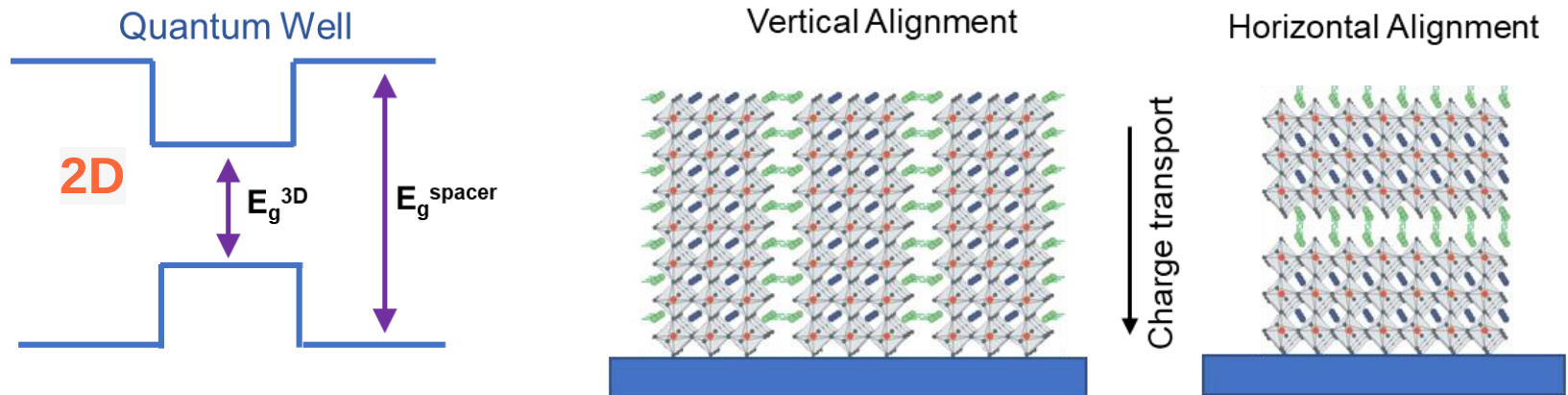
Why 2D structure is more stable than 3D structure?

- Strong hydrogen bonding between the spacer cation and inorganic octahedron
- Hydrophobic property of organic spacer cation

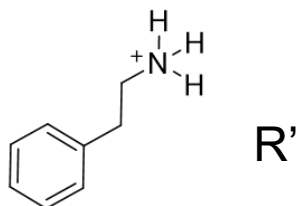


Disadvantages of 2D Perovskite

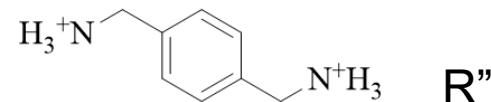
- Quantum well structure is not favorable for charge separation
- Vertical orientation is required for efficient charge transport because charge transport through organic layer is poor.
- Therefore, efficiency of 2D perovskite is poor although the stability is much improved.



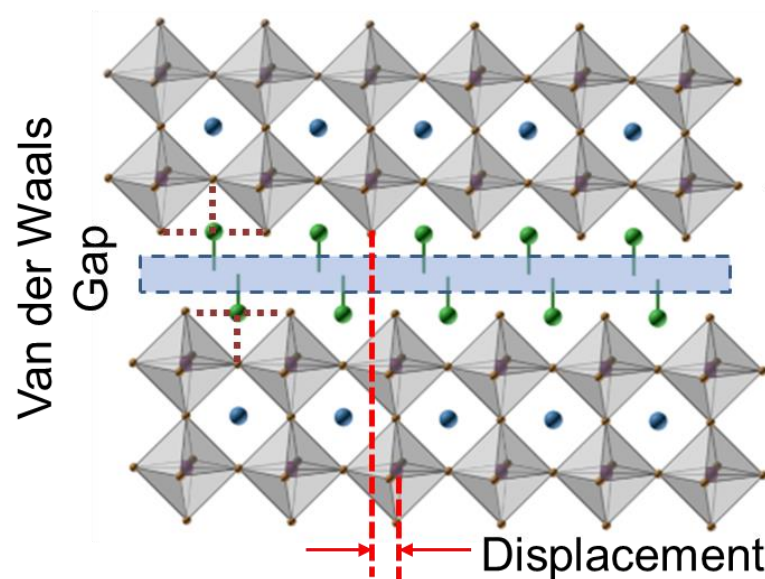
Ruddlesden-Popper (RP) Phase Vs. Dion-Jacobson (DJ) phase



2 monovalent cations

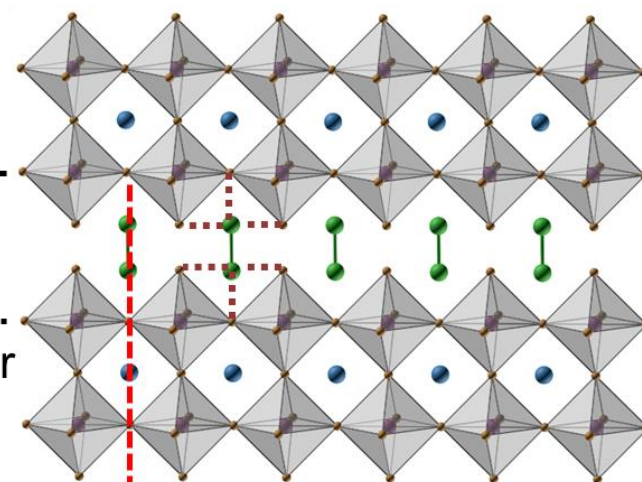


1 divalent cation



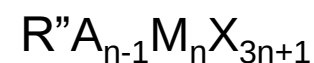
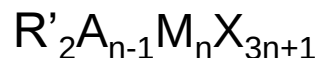
R-P Phase

Interlayer distance



D-J Phase

..... Hydrogen bond



Advantages of DJ Phase over RP Phase

- DJ phase has advantage in structural stability and lattice stiffness
 - ❑ the alternate hydrogen bond interactions between the organic spacer and inorganic slab without any Van der Waals gap
 - ❑ a short interlayer distance between inorganic slabs
 - ❑ a stack with perfect alignment or minor displacement between unit cells
- In RP phase:
 - ❑ only a weak Van der Waals force is holding between layers
 - ❑ imperfect stacking alignment between unit cells ($\frac{1}{2}$, $\frac{1}{2}$ displacement) due to two monovalent spacers in interdigitated form
- In addition to structural stability, the closely packed inorganic slabs in DJ phase also weakens the thermally included electron-phonon scattering and enhances the delocalization of band-edge charge densities, resulting in longer carrier lifetime